B VISHNU SAI

+91 9880869432 | vishnusai0317@gmail.com | https://www.linkedin.com/in/vishnusai0317

Summary

Computer Science Engineering student with strong foundations in C programming, embedded systems, and hardware interfacing. Experienced in developing firmware for IoT devices using Embedded C, working with sensors, microcontrollers, and real-time data processing. Familiar with hardware communication protocols and system-level debugging. Passionate about low-level programming, firmware development, and building reliable hardware-software integrated systems.

Skills

Programming**:** C (Embedded Systems), C++, Python

Embedded Systems: Embedded C, Microcontroller Programming (ESP8266 / NodeMCU), Firmware Development, Sensor Interfacing, Real-Time Monitoring Systems, Hardware-Software Integration, Embedded Debugging

Communication Protocols**:** I2C**,** SPI**,** UART

Hardware Knowledge**:** Microcontrollers & Embedded Systems**,** CPU Architecture Basics**,** Memory Systems (RAM / Flash)**,** Hardware Debugging**,** IoT Hardware Integration, Interrupt Handling

Tools & Platforms**:** Git, VS Code, IntelliJ IDEA, IoT Development Kits

Projects

Early Warning System for Glacial Lake Outburst Floods | NodeMCU (ESP8266), Embedded C, IoT Sensors

- Developed **embedded firmware in C** for the ESP8266 microcontroller to process real-time sensor data.
- Implemented a **multi-sensor data acquisition system** for water level, vibration, temperature, and GPS sensors.
- Designed **threshold-based interrupt alerts** for early flood detection.
- Integrated sensors using **hardware communication interfaces (I2C / GPIO)**.
- Optimised sensor polling and alert logic for **real-time response and reliability**.
- Implemented remote monitoring and alert notifications using the Blynk IoT dashboard.

I2C Temperature Sensor Interface | Embedded C, ESP8266, I2C Protocol

- Developed firmware to interface an I2C temperature sensor with ESP8266 microcontroller.
- Implemented I2C communication protocol for reliable data transfer between microcontroller and temperature sensor.
- Processed raw sensor data and converted it into readable temperature values.
- Displayed real-time temperature data via serial monitoring for debugging and validation.

Education

Bachelor of Engineering (B.E.) – Computer Science Engineering — Nov 2022 - Present
Rao Bahadur Y. Mahabaleswarappa Engineering College, Ballari — *CGPA: 8.02 / 10*

Awards & Certifications

- Certified in C Programming
- Workshop on IoT

Relevant Coursework

Data Structures, Computer Organisation & Architecture, Embedded Systems, Computer Networks, Operating Systems